

Hassan S

Backend and Gen AI Engineer. Built VentureGPT for venture studios, implementing automated multi-agent workflows. Looking to build production-grade AI systems at a frontier AI company.

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EXPERIENCE

SDE & AI/ML Engineer

2025 – Present

HTC Global Services

- Migrated the company's legacy timesheet application to a modern Java full stack by implementing **JWT authentication** and **role-based access control** on **Spring Boot and React.js**.
- Brought OCR for 19th-century handwritten manuscripts in-house after the company had been outsourcing it, building a hybrid pipeline (**Kraken** line segmentation + **Microsoft TrOCR**) for English and fine-tuning a separate **Chinese open-weight model** for Chinese manuscripts, validated through a **Gemini Vision API / CER-WER** evaluation framework.
- Eliminated a fully manual, paragraph-by-paragraph bounding-box workflow for newspaper text extraction by fine-tuning a **YOLO object detection model**, reaching **0.95 accuracy**.
- Operate production backend services by building **REST APIs** and **microservices** in **Java and Spring Boot**, handling deployments, data integrations, and production support across the team's stack.

Software Developer

2024 – 2025

Aziwa Infosys Systems, Chennai

- Shipped **REST APIs** in a Java and Spring production system supporting CRUD operations and client workflows.
- Deployed services to **AWS EC2**, wrote cron jobs for recurring data tasks, and tested APIs with Postman.
- Refactored a legacy Java codebase and built a **Salesforce AppExchange integration**.

PROJECTS

VentureGPT

Freelance · Venture studio

Multi-agent VC due-diligence platform · Python, LangGraph, FastAPI, PostgreSQL, Redis

- Built an API-first platform orchestrating **7 specialist AI agents** (company, founder, competitor, market, technical, risk, memo) through a **LangGraph** workflow with parallel fan-out, producing evidence-grounded investment scores, a memo, and an advisory Go/No-Go.
- Designed a typed shared-state model with **reducer-based parallel merging** and a **human-in-the-loop approval gate**, backed by versioned Postgres checkpoints; built a ports-and-adapters tool layer enabling **148 deterministic offline tests** and zero-code-change deployment.

Enterprise RAG

Solo · Internal R&D

Hybrid semantic + keyword search for enterprise Q&A

- Built a production-grade RAG pipeline as independent R&D and introduced it internally. The team adopted it after validating the results, built solo end-to-end.
- Vector search powered by **HNSW (Hierarchical Navigable Small World)** indexing in **Qdrant**, combining semantic similarity with keyword search for **hybrid retrieval**.

EDUCATION

B.E. Computer Science and Engineering

Mohamed Sathak A.J. College of Engineering, Chennai

SKILLS

Languages Python, Java, SQL, JavaScript

Gen AI LLM fine-tuning (open-weight models), multi-agent systems (LangGraph), RAG (Qdrant, HNSW, hybrid search), OCR & computer vision (YOLO, Kraken, TrOCR)

Backend Spring Boot, FastAPI, REST APIs, Microservices, PostgreSQL, Redis, MySQL

AI Coding Agents Claude Code, Cursor, parallel multi-agent coding workflows

Infrastructure AWS EC2, Docker, Git, Postman, JWT & role-based access control

Certification Microsoft Azure AI-102